

MANUEL LANCASTRE

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
M.S. in Artificial Intelligence Engineering

Carnegie Mellon University Pittsburgh, PA
B.S. in Mechanical Engineering, University Honors, GPA: 3.5

EXPERIENCE

Software Engineer Aug. 2025 – Present
Oracle (Opower) Boston, MA

- Shipped the **AMI Publisher**, streaming usage data in **Apache Iceberg** format for **millions of meters** across utility partners, with an open-source consumption guide and automated performance testing to match
- Building an **MCP server** that exposes disaggregation insights and billing/account data as tools for a customer-facing conversational agent that answers live billing questions
- Built an autonomous **on-call agent** (Codex SDK) that triages and resolves incidents end to end, cutting manual on-call time ~ **80%**
- Own the deep-learning pipelines powering Opower's per-appliance energy disaggregation and led the expansion of coverage to **4.6M business sites**, alongside **45M+ households** served
- AI Ambassador**: built internal Codex skills and plugins adopted by **100s of engineers**, saving **1000s of hours** org-wide

PROJECTS

Adaptive Prosthesis Control via Deep Imitation Learning | *Python, PyTorch, MuJoCo, PPO, VAIL* 2025

- Pioneered a two-agent imitation-learning framework using Variational Adversarial Imitation Learning (VAIL) for biomechanically-accurate transtibial prosthesis control, achieving superior gait mimicry over traditional finite-state-machine approaches
- Ran a comparative study of imitation learning vs. deep RL (PPO) for ankle-level assistive control; IL produced **3x longer** stable locomotion in physics-informed MuJoCo simulations
- Engineered adaptive control policies with progressive sensor limitation ($36 \rightarrow 22 \rightarrow 16$ state observations), keeping robust performance despite a **55% reduction** in input dimensionality
- Built a sparse-reward PPO pipeline that achieved emergent locomotion from minimal shaping (+1 per timestep survival, -100 terminal), demonstrating sample-efficient policy learning without expert demonstrations
- Developed an MLP behavior-cloning ankle controller minimizing action prediction error against expert VAIL demos, enabling user-adaptive prosthetic control across diverse locomotion conditions
- Published project site and paper

Nutritio: Edge-First AI Nutrition Platform | *Python, PyTorch, Transformers, React* 2025

- Built Nutritio into a "ChatGPT for food": a single conversational agent (**Kimi-k2.5**) that lets users track meals, plan groceries, and get restaurant picks just by talking to it, instead of juggling separate tracking apps
- Designed the agent's tool-calling architecture, spanning **20+ tools** across meal/macro tracking, pantry and receipt intake, recipe planning, and live restaurant search, all backed by persistent SQLite/Supabase state that gives the agent durable memory of a user's pantry, goals, and history across conversations
- Fine-tuned a custom **Gemma 3-1B** architecture with regression heads for multi-target nutrient prediction, achieving a **4x** accuracy improvement over zero-shot baseline
- Built a live restaurant-recommendation pipeline (web discovery + menu extraction) that filters real menus by dietary goals and allergens, validated with an automated eval against real listings

TECHNICAL SKILLS

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- Languages**: Python, Java, C/C++, SQL, JavaScript/TypeScript, R, MATLAB
 - ML / AI**: PyTorch, TensorFlow, JAX, Hugging Face, MuJoCo, Apache Iceberg, Pandas, NumPy, LangGraph
 - Agents & LLMs**: Codex SDK, MCP, Anthropic API, prompt engineering, agent evaluation, behavior cloning
 - Frameworks & Tools**: React, FastAPI, Node.js, Docker, Kubernetes, AWS, OCI, Grafana, Splunk, Git